RAVINDRA YADAV

AI & Full-Stack Developer

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PROFESSIONAL SUMMARY

AI & Full-Stack Developer building practical web applications and AI-powered products. Skilled in full-stack development, backend systems, and LLM integrations, including multi-LLM platforms, RAG pipelines, and privacy-focused AI tools. Focused on improving performance, designing scalable systems, and delivering reliable user experiences.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, C, C++

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS, Bootstrap

Backend: Node.js, Express.js, REST APIs, MVC Architecture, JWT, WebSockets, Rate Limiting

AI / GenAI: OpenAI API, Gemini API, Vercel AI SDK, Tool Calling, Streaming, Prompt Engineering, RAG Fundamentals, Embeddings, Multi-LLM

Databases & Cloud: MongoDB, PostgreSQL, MySQL, Mongoose, Drizzle ORM, Docker, Git, Redis, Upstash, Vercel, Render, R2 Storage

CS & Networking: Operating Systems, Computer Networks, OOP, DBMS, System Design (HLD)

PROJECTS

SkillFarm - AI Learning Platform : Demo

Next.js • React • TypeScript • PostgreSQL • Drizzle ORM • Auth.js • Redis • OpenAI • Gemini

- Built an AI learning workspace with specialized mentors across six technical domains, curated learning resources, and personalized roadmaps.
- Optimized mentor response times through prompt optimization and reuse across specialized mentors, reducing latency and additional token usage.
- Developed an admin monitoring dashboard to track per-user LLM costs and response latency, enabling usage analysis and performance optimization across specialized mentors.

P2Shield - Client-Side LLM Privacy Firewall & Token Vault : Demo

TypeScript • React • Next.js • Web Extensions API • NLP/NER • Web Crypto API

- Built an on-device privacy firewall combining regex and NER detection with context-preserving synthetic surrogates to protect sensitive data in browser-based AI chats.
- Implemented an AES-GCM 256-bit encrypted token vault with automatic bidirectional unmasking and a local monitoring dashboard for privacy risks and token lifecycle tracking.
- Integrated privacy controls directly into dynamic browser chat interfaces, supporting real-time interception and protection of prompts without disrupting the user experience.

ChatMate - AI-Powered Multi-LLM Chat Platform : Demo

React • Next.js • TypeScript • Node.js • MongoDB • OpenAI • Gemini • AI SDK • Tailwind CSS

- Engineered a multi-LLM chat platform with real-time streaming, modular AI architecture, and tool calling.
- Implemented conversation branching with preserved history and reusable React components for a responsive chat interface.

ChaiBookLM - AI-Powered Research Assistant : Demo

Next.js • React • TypeScript • PostgreSQL • Prisma • Qdrant • OpenAI • Gemini • Tailwind CSS

- Built a RAG research assistant supporting PDFs, websites, YouTube videos, and transcripts with grounded, clickable source citations.
- Engineered a modular ingestion and retrieval pipeline using source-specific parsers, chunking, embeddings, Qdrant vector search, query rewriting, and isolated notebook knowledge bases.

ACHIEVEMENTS & CERTIFICATIONS

- **First Prize, Martinovation 2026** - BitShift Hackathon | Team TechKnights
- **GenAI with JavaScript 2026** - Certificate of Completion | ChaiCode

TECHNICAL WRITING

Technical Content Author: Published technical articles simplifying concepts including

- Where RAG Fails: Understanding the Limitations ↗
- Context Window — Humans vs. LLMs: How They Remember Information ↗
- How ChatGPT Understands Your Questions: From Text to Intelligence ↗

EDUCATION

Bachelor of Computer Applications (BCA) — Dr. Shyama Prasad Mukherjee University, Ranchi | CGPA: 9.1 (2024 – 2027)

Senior Secondary (Class XII) — St. Xavier's College, Ranchi | JAC — 87% (2024)